

AKSHAYA AYYASAMY

THIRAVIDA MANI

Java Backend Developer | Agentic AI Systems | B.Tech IT

SUMMARY

- Backend Developer focused on building high-performance, concurrent Java systems and integrating Agentic AI capabilities (RAG, tool-calling) via Spring Boot, Spring AI, and LangChain4j.
- Completed Zoho Schools' Marupadi program, mastering full-stack Java development, Advanced DBMS, and Low-Level Design (LLD).
- Prior experience as an iOS Developer at Cognizant; transitioning back into engineering with a sharp focus on modern backend architecture and autonomous LLM integrations.

PROJECTS

Employee Leave Management System (ELMS) — Agentic RAG & MCP
Intelligent leave management platform replacing traditional forms with an AI conversational assistant.

- Architected a Spring Boot backend bridging a conversational AI interface with real-world database transactions via the Model Context Protocol (MCP).
- Implemented Agentic RAG using pgvector to allow the AI to accurately answer company policy questions based on embedded HR documents.
- Enforced strict pessimistic database locking to ensure leave balances are never double-spent during concurrent AI tool-calling requests.

AI Outlast – Multi-Agent LLM Simulation *Autonomous LLM multi-agent simulation where AI personas compete, form alliances, and vote.*

- Engineered a robust multi-agent orchestration engine in Java 17 and Spring Boot 4 supporting asymmetric social memory matrices.
- Integrated a provider-agnostic LLM routing layer (Groq, Gemini, Ollama) behind a REST API.
- Implemented probabilistic deception mechanics and real-time frontend polling to visualize the AI agents' internal reasoning and trust networks.

Lift Management System *Highly concurrent elevator simulation demonstrating deep understanding of Java multithreading.*

- Simulated a multi-lift system utilizing one worker thread per active elevator.
- Implemented a SCAN-style dispatch algorithm using separate thread-safe `PriorityBlockingQueues` for directional stops.
- Orchestrated complex thread coordination (hall-call → boarding → car-call) using raw `wait()`/`notifyAll()` monitor synchronization.

WORK EXPERIENCE & TRAINING

Zoho Schools – Marupadi | *Software Development Trainee* (Jan 2026 – Jun 2026) *Intensive software engineering program focused on production-grade Java and data structures.*

- Mastered object-oriented programming, concurrent data structures, and Low-Level Design (LLD) in Java.
- Designed normalized database schemas and implemented full-stack web applications using HTML, CSS, JavaScript, and Java backends.

Cognizant Technology Solutions | *Programmer Analyst – iOS Developer* (Dec 2009 – Jul 2011) *Global IT services and consulting company.*

- Developed native iPad applications on iOS 3, shipping a touch-based signature input module used in production.
- Implemented drag-and-drop inventory placement features and complex user interaction workflows.
- Collaborated closely with Java backend teams to design DBMS tables and integrate mobile clients with backend REST services.

SKILLS

- **Backend & Systems:** Java (11-21), Spring Boot, Java Servlets, REST APIs, Multithreading, Low-Level Design (LLD), Concurrent Data Structures.
- **AI & Agentic Systems:** Spring AI, LangChain4j, Multi-Agent Systems, Agentic RAG, MCP Integration, LLM APIs (Gemini/Ollama).
- **Databases & Tools:** PostgreSQL, pgvector, Docker, Git, Maven, Flyway, JWT Auth.

EDUCATION

B.Tech – Information Technology, College of Engineering, Guindy (CEG), Anna University

ACHIEVEMENTS

- **UPSC Civil Services:** Cleared the highly competitive UPSC Preliminary Examination thrice (2015, 2016, 2017), demonstrating extreme discipline, analytical thinking, and persistence.